

Logical operators binary truth table definition

Input		Output				
p	q	Conjunction $p \wedge q$	Exclusive or $p \oplus q$	Disjunction $p \vee q$	Conditional $p \rightarrow q$	Biconditional $p \leftrightarrow q$
T	T	T	F	T	T	T
T	F	F	T	T	F	F
F	T	F	T	T	T	F
F	F	F	F	F	T	T
		“ p and q ”	“ p xor q ”	“ p or q ”	“if p then q ”	“ p if and only if q ”

Logical operators truth tables

Truth tables: Input-output tables where we use T for 1 and F for 0.

Input		Output		
p	q	Conjunction $p \wedge q$	Exclusive or $p \oplus q$	Disjunction $p \vee q$
T	T	T	F	T
T	F	F	T	T
F	T	F	T	T
F	F	F	F	F
				

Input	Output
p	$\neg p$
T	F
F	T
	